VOYAGEHACK 2.0

GuardianX

Problem Statement: Fraud Prevention
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Solution, System Architecture & Decision Framework

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This solution is built as a real-time, intelligence-driven architecture that evaluates every agency interaction through risk, trust, and business context. Instead of fixed rules, it uses a three-layer decision system that continuously learns and adapts.

Layer 1: Real-Time Signal Processing

- **Behavioral:** Login consistency, booking spikes, and risky cancellation behavior
- **Financial:** Credit usage trends, payment discipline, and exposure vs revenue balance
- **Contextual:** Agency maturity, seasonal demand patterns, and network linkages

Layer 2: Composite Risk Scoring

Trust Score = Base × Behavior × Financial Health × Anomaly Discount

Risk Bands

- **90-100**: Green (auto-approve, credit expansion eligible)
- **70-89**: Yellow (standard monitoring, holds on anomalies)
- **50-69**: Orange (enhanced review, reduced limits)
- **<50**: Red (manual approval required, possible suspension)

Layer 3: Adaptive Action Engine

Credit Limit Decisions: 1.Contract: Trust score drops >15 points in 30 days OR utilization >80% for 14+ days

2. Expand: Trust score >85 for 90+ days + <60% utilization + zero disputes

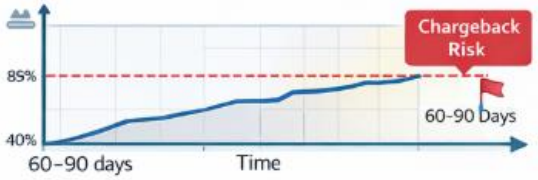
3. Pause: Anomaly detected (geo-velocity violation, device switch during high-value booking)

Key Innovation: Temporal Pattern Recognition

Chargeback Prediction Model

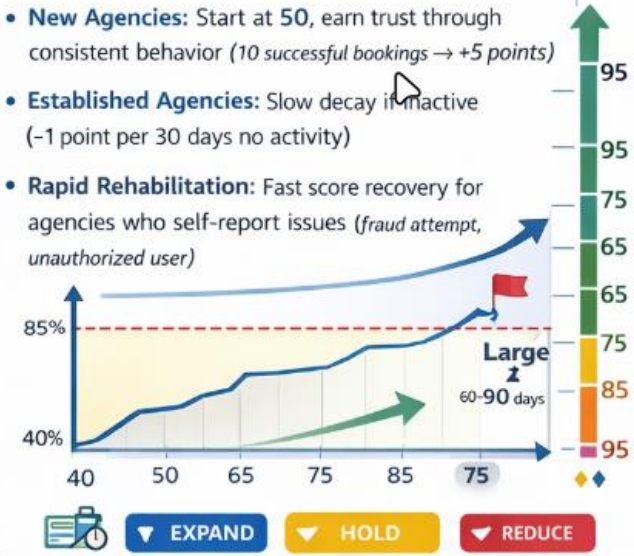
Early warning signals (60-90 days before chargbock:

1. **Creeping Utilization:** Gradual increase from **40%** → **85%+** over 8 weeks
2. **Booking Profile Shift:** Shift from domestic to international, or economy to premium
3. **Communication Degradation:** Slower response times to invoices / queries
4. **Micro-Cancellations:** Small test cancellations before large disputed bookings



Trust Evolution Tracking

- **New Agencies:** Start at **50**, earn trust through consistent behavior *(10 successful bookings → +5 points)*
- **Established Agencies:** Slow decay if inactive (-1 point per 30 days no activity)
- **Rapid Rehabilitation:** Fast score recovery for agencies who self-report issues *(fraud attempt, unauthorized user)*



Tech Stack: Python + FastAPI + PostgreSQL + Redis + Kafka , PyTorch + LangChain + Claude API , Docker + K8s + AWS

Implementation Strategy & Outcomes

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Agent 1: Anomaly Detector

Responsibility : Continuous monitoring of 50+ signals per agency.

- Detects deviations >2 standard deviations from baseline.
- Clusters related anomalies.
- Generates "investigation prompts" for Agent 2 .

Agent 2: Contextual Reasoner

Responsibility: Interprets anomalies with business context

- Checks for legitimate explanations (seasonal events)
- Cross-references with external data
- Escalates only unresolved high-confidence risks

Decision Tree Logic:

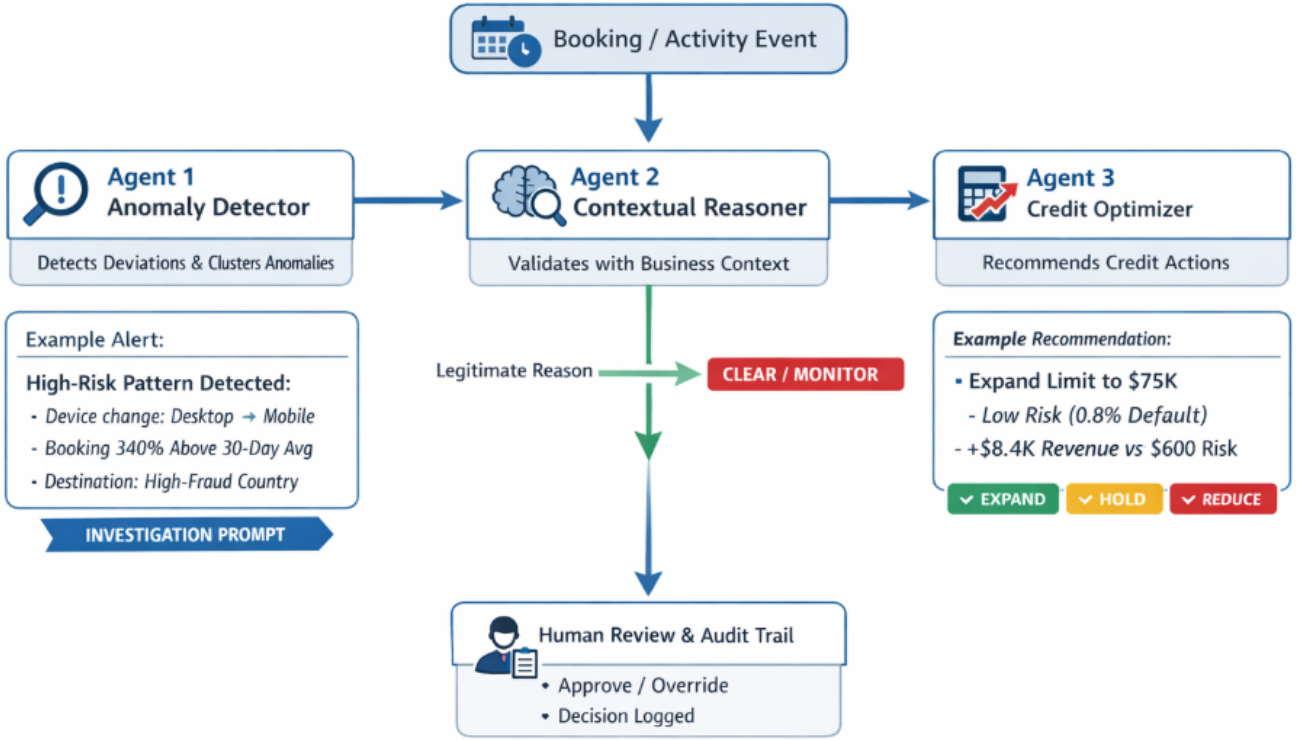
- **Geo Anomalies:** Clear if multi-office + destination matches; otherwise escalate with confidence score.
- **Utilization Spikes:** Monitor if seasonal/planned; otherwise flag for proactive risk outreach.

Agent 3: Credit Optimizer

Responsibility: Proactive credit limit adjustments

- Weekly batch review of all agencies in Yellow/Orange bands
- Simulates credit expansion impact (revenue gain vs. risk exposure)
- Recommends human-in-the-loop decisions with full reasoning

Agentic Workflow: The "Credit Guardian" System



Outcome: 40% fewer chargebacks, 60% fraud loss, false positives from 15% → 6%, manual reviews from 22% → 5%, credit utilization by ~30%.

Explainability & Human Oversight: (Decision Audit Trail) : Every action logged with: Input signals and values , Decision logic path taken ,Confidence score and alternative options considered, Human override option with reason capture

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THANK YOU